

Power BI Project Documentation

Sales & Customer Insights Dashboard

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1. Project Overview

This project is designed to analyze and monitor business performance across sales, profit, customers, returns, and shipping operations. The dashboard provides decision-makers with a centralized, interactive view of key metrics.

Objectives

- Track overall business performance (Sales, Profit, Orders)
- Understand customer behavior and segmentation
- Analyze regional performance
- Monitor returns and shipping efficiency
- Identify trends and growth opportunities

2. Data Model Overview

Schema Type: Star Schema

The data model follows a **star schema design**, ensuring:

- High performance
- Easy filtering and slicing
- Clear separation of fact and dimension tables

Tables Used

Fact Table

Orders Table (Central Table)

Contains transactional data:

- Order ID
- Order Date
- Customer ID
- Sales, Profit, Quantity
- Discount, Cost, Net Price
- Product & Geography details

❖ This table drives all calculations and aggregations.

Dimension Tables

1. People Table

- Person
- Region

Used for mapping sales ownership and regional analysis

2. Returns Table

- Order ID
- Returned (Yes/No)
- Return Rate %
- Total Returns

Used to analyze return behavior and product issues

Relationships

- Orders → People (Region-based relationship)
- Orders → Returns (Order ID relationship)

Filter Direction

- **Single Direction Filtering**
- From dimension → fact
- Improves performance and avoids ambiguity

3. Data Preparation

Derived Columns Created

- Price after Discount
- Cost Price Calculations
- Customer-related attributes
- Region/Category groupings

DAX Measures Created

- Total Sales - Total Sales = `SUM(Orders[Netprice])`
- Total Profit - Total Profit = `SUM(Orders[profit amt])`
- Total Orders - Total Orders = `COUNT(Orders[Order ID])`
- Total Customers - Total Customers = `DISTINCTCOUNT(Orders[Customer ID])`
- Profit Margin % - Profit Margin% = `DIVIDE(SUM(Orders[profit amt]), SUM(Orders[Netprice])) * 100`
- YoY Growth - YOY Growth% = `DIVIDE([YOY Sales Growth],[Sales LY])`
- YoY Sales Growth - `TOTALYTD([Total Sales], Orders[Order Date])`
- Return Rate % - ReturnRate% = `DIVIDE([Total Returns], [Total Orders])`

4. Key Metrics (KPIs)

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin %
- Return Rate
- Year-over-Year Growth

These KPIs provide a quick executive summary of business performance.

5. Dashboard Pages

1. Executive Overview

Purpose: High-level performance snapshot

Key Visuals:

- KPI Cards (Sales, Profit, Orders, Customers)
- Trend Line (Sales over time)
- YoY Growth indicators

Helps leadership quickly assess performance.

2. Sales & Profit Analysis

Purpose: Deep dive into revenue and profitability

Key Insights:

- Sales by Category & Sub-category
- Profit vs Sales comparison
- High-performing products

Identifies profitable and loss-making areas.

3. Geography Analysis

Purpose: Regional performance tracking

Key Visuals:

- Sales by Region/State
- Map visualization
- Region-wise profitability

Helps identify strong and weak markets.

4. Segment Analysis

Purpose: Customer segmentation insights

Segments Used:

- Consumer
- Corporate
- Home Office

Insights:

- Sales by segment
- Profit contribution
- Customer behavior patterns

5. Returns Analysis

Purpose: Monitor product returns

Key Insights:

- Return Rate %
- Returned Orders
- Impact on profit

Helps identify product or logistics issues.

6. Shipping Analysis

Purpose: Evaluate delivery performance

Key Insights:

- Ship Mode usage
- Delivery trends
- Cost vs efficiency

Helps optimize logistics strategy.

6. Visualizations Used

- KPI Cards
- Line Charts (Trend Analysis)
- Bar Charts (Comparisons)
- Map Visuals (Geography)
- Matrix (Heatmaps with conditional formatting)
- Scatter Charts (Correlation analysis)

7. Business Insights Delivered

- Identification of high-revenue regions
- Understanding of customer segments driving growth
- Detection of high return-rate products
- Profitability gaps across categories
- Shipping inefficiencies

8. Key Features

- Interactive filtering (Date, Region, Segment)
- Drill-down capabilities
- Clean and user-friendly layout
- Real-time insights

9. Challenges & Solutions

Challenge	Solution
Data inconsistency	Data cleaning & transformation
Complex calculations	Efficient DAX measures
Performance issues	Star schema + single filter direction
Return tracking	Integrated Returns table

10. Conclusion

This dashboard provides a comprehensive 360° view of business performance. It enables stakeholders to:

- Make data-driven decisions
- Improve profitability
- Optimize operations
- Enhance customer satisfaction

11. Future Enhancements

- Predictive analytics (forecasting)
- Customer churn analysis
- Inventory optimization
- Advanced drill-through pages